RealQM for H_2 and He

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Abstract

RealQM models the ground state of an atomic system with kernels Z_a at positions $\mathbf{R}_a$ surrounded by N electrons in terms of a wave function $\Psi = \sum_i \psi_i$, a single scalar function on $\mathbb{R}^3$ in H^1, with ψ_i supported on a domain Ω_i, the domains disjoint and filling space and each carrying unit charge, minimising the energy over both Ψ and the partition.

For $N = 1$ this is the standard model for hydrogen, calibrated there by the coefficient $\frac{1}{2}$ of the kinetic energy against the observed spectrum, which makes $N = 1$ exact. The step that makes RealQM new is $N = 2$, from H to H_2, and this article takes that step—without any further calibration, and in the two versions the interface between domains admits.

1 The model

RealQM models the ground state of an atomic system—kernels of charge Z_a at positions $\mathbf{R}_a$, surrounded by N electrons—in terms of a wave function

$$\Psi \;=\; \sum_{i=1}^{N} \psi_i \;\in\; H^1(\mathbb{R}^3), \qquad \int_{\Omega_i} \Psi^2 = 1 \;\text{ for each } i, \tag{1}$$

in which $\psi_i = \Psi|_{\Omega_i}$ is the restriction of Ψ to a domain Ω_i, the Ω_i being disjoint and filling $\mathbb{R}^3$, as the minimiser of the energy

$$E[\Psi, \{\Omega_i\}] \;=\; \underbrace{\tfrac{1}{2}\int |\nabla\Psi|^2}_{\text{kinetic}} \;-\; \underbrace{\int K\,\Psi^2}_{\text{electron–kernel}} \;+\; \underbrace{\sum_{i<j} \iint \frac{\psi_i^2(\mathbf{x})\,\psi_j^2(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|}}_{\text{electron–electron}} \;+\; \underbrace{\sum_{a<b} \frac{Z_a Z_b}{R_{ab}}}_{\text{kernel–kernel}}, \tag{2}$$

with $K = \sum_a Z_a/|\mathbf{x} - \mathbf{R}_a|$ the kernel attraction, the minimisation being over *both* Ψ and the partition $\{\Omega_i\}$—the domains are unknowns of the problem, not given data. The only inputs are the kernel charges, their positions, and N. The domains being disjoint, $\Psi^2 = \sum_i \psi_i^2$ with no cross terms, so the third sum is a genuine pair interaction counted once, and no electron interacts with itself.

For $N = 1$ this is the standard model for hydrogen. With one electron there is one domain, no partition to determine and no interface, and (2) reduces to $\frac{1}{2}\int |\nabla\Psi|^2 - \int (Z/r)\Psi^2$ minimised subject to $\int \Psi^2 = 1$. Nothing distinguishes the construction from the textbook one here: the two are the same variational problem.

The critical step is $N = 2$. What makes RealQM fundamentally different from textbook quantum mechanics is the extension from one electron to two—from H to H_2—since that is where the partition, the interface between domains, and the question of what holds two electrons apart all appear for the first time. This article isolates exactly that step. The case $N > 2$, where several heavy kernels and core shells enter, is taken up in the companion article

on the molecule simulator [2], which carries sixteen molecules built from H, C, N and O on the same declared mesh.

Three features of the statement above are worth drawing out, because the rest of the paper turns on them.

Continuity is not imposed; it is the function space. Ψ is required to lie in H^1, so it is continuous across every interface by assumption, and no matching condition is added by hand. What *is* free is where the interfaces are.

The interface condition is a consequence, not an input. Varying Ψ at a *fixed* partition leaves a boundary term $\oint \psi_i \, \partial\psi_i/\partial n$ on each $\partial\Omega_i$, so the natural condition is $\partial\Psi/\partial n = 0$ from each side—nothing is imposed. Solved that way the two one-sided values at an interface generically disagree, and the interface is then placed where they agree, which is what varying the *partition* supplies. Three conditions, one extra unknown; the count closes. The two implementations in §3 are two discretisations of this one variational problem and differ only in how they represent Ψ near the interfaces—not in the problem being solved.

There is one function, not N of them, and it is a scalar function in three dimensions. The decomposition (1) is bookkeeping: ψ_i is a restriction of Ψ, not an independent orbital, and the Ω_i are the only thing distinguishing one electron from another. So the unknown is a single scalar field on $\mathbb{R}^3$ whatever N may be, and at a spatial resolution h it is represented by a mesh of h^{-3} points on a unit cube—some 10^3 values per a_0^3 at the $h = 0.1\,a_0$ used here, independent of the number of electrons.

That is the whole of why this model is computable, and it is worth stating as the arithmetic rather than as a claim. The cost of a calculation grows with N only through the number of domains to be tracked and the pair sum in (2), not through the dimension of the unknown, so adding an electron adds work linearly rather than multiplying the size of the space to be searched. Every number in this paper was produced in a web browser [1], which would not be possible otherwise.

Two mechanisms, doing two different jobs. It is worth separating them, because they are easily run together. The Coulomb term does *not* keep two electrons apart: for smooth densities $\iint \rho(\mathbf{x})\rho(\mathbf{y})/|\mathbf{x}-\mathbf{y}|$ is finite even when the two coincide, so it permits coincidence at a finite price and cannot forbid it. What forbids it is the *partition*, which is a constraint and not a penalty: the domains are disjoint by construction, so overlap is unavailable rather than expensive. What the Coulomb term does is resist *compression*—as two domains are pushed together V_{ee} rises, and that sets where the interface sits and how density distributes inside each domain.

The distinction matters for what follows. Between *equivalent* domains across a mirror plane the partition contributes nothing beyond what Coulomb already supplies: the hard and weak constructions of §3 agree there to 4×10^{-8}. So in exactly that configuration the construction's whole resistance to compression is the classical Coulomb repulsion of two charge clouds, and the over-binding reported in §5 is the measurement that this alone is not enough. H_2 at its midplane and He at its split plane are both that configuration, which is why they are the systems to examine.

What is new, and what is missing. Relative to one electron the construction adds exactly two things: the densities do not overlap, and they must meet somewhere. It also *subtracts* one: it fixes what resists their approach, and the answer is set out below—the partition makes overlap impossible while the Coulomb term alone resists compression. That is a property of

the model, stated in its own terms and checkable within it, and it is why no two-electron system should be expected to come out exactly.

2 One calibration, on hydrogen

The coefficient $\frac{1}{2}$ on the kinetic term in (2) is the model's calibration, and hydrogen is what sets it. A kinetic penalty quadratic in $\nabla\Psi$ is the lowest-order rotation-invariant term that prevents collapse, so the *form* is forced; its coefficient is a scale, and the construction contains nothing from which to derive one—no mass appears anywhere in (2) and no constant of action. Fixing it against the observed spectrum of hydrogen gives $\frac{1}{2}$ in atomic units, whereupon the minimiser is $\Psi = \sqrt{Z^3/\pi}\, e^{-Zr}$ with energy $-Z^2/2$: **the model for $N = 1$ is thereby exact**, and exact by calibration rather than by prediction. That is why no agreement at one electron can be evidence for the construction, and why the critical step is $N = 2$.

The continuum model is then complete and contains no further adjustable quantity. What remains is a second calibration of an entirely different kind, belonging to the discrete scheme rather than to the model.

One mesh for all N. We fix

$$h = 0.1\, a_0, \qquad \text{box } 12\, a_0, \qquad 120^3 \text{ cells}, \tag{3}$$

and use it for every system regardless of N. The mesh is not adjusted per system and no mesh ladder is run to justify a number.

Calibrate the discrete model to the exact model. A grid cannot represent $1/r$, so the kernel carries a softening, $K = Z/\sqrt{r^2 + \varepsilon^2}$ with $\varepsilon = \sigma h$. We fix σ by requiring the *discrete* model to reproduce the *exact* model's one-electron energy,

$$E_h(\mathrm{H}) \;=\; -\tfrac{1}{2} \qquad \Longrightarrow \qquad \sigma = 0.627 \;\text{ at } h = 0.1, \tag{4}$$

from a four-point ladder (-0.508456, -0.500962, -0.493833, -0.486591 at $\sigma = 0.4, 0.6, 0.8, 1.0$, linear in σ to better than a percent). Verified rather than extrapolated: at (3) it returns -0.499854, an error of 0.03%, and the same σ gives -0.500067 at box $16\, a_0$ and -0.499901 at box 20—two above $-\frac{1}{2}$ and one below, all within 2×10^{-4}. So the calibration is box-independent at the level the scheme assumes, and the residual is scatter rather than a trend.

Note what this calibration is *against*. Not experiment: the target $-\frac{1}{2}$ is the exact model's own value, already fixed by the kinetic coefficient of §1. So σ is a discretisation constant, chosen so that the computed scheme reproduces the continuum problem it is meant to solve, and it consumes no observable at all. Strictly it is $\sigma(h)$—a different mesh requires re-solving (4) rather than reusing 0.627—and since less softening is needed as the grid improves, $\varepsilon = \sigma(h)\, h \to 0$: no finite softening is frozen into the continuum limit.

The ledger, in full. One coefficient calibrated against observation (the $\frac{1}{2}$ of §1, on hydrogen's spectrum, which makes the exact model exact at $N = 1$); one mesh declared (3); one softening determined (4), against the exact model and not against data. Nothing is fixed on a two-electron system, which is what makes everything below a prediction.

Where the mesh comes from, and what precision to expect. Neither number is chosen on a physical ground. The spacing is the finest at which these systems compute in a *common web browser*, and the box the smallest that holds them: every system here has two electrons on one or two kernels, so none needs room for a polyatomic cloud, and $12\, a_0$ suffices. The

cost is steeply non-linear in the box—120^3 runs to convergence in minutes where 160^3 takes roughly four times as long per step and 200^3 exhausts the browser's resources and returns no energy at all—so the box is set by what computes, not by what would be comfortable. The companion article, which carries polyatomics, declares a larger box for that reason and pays for it. The whole simulator is deliberately a browser program [1]—anyone can load it and reproduce every number here without installing anything, each row being specified by a query string—and $h = 0.1$ is the resolution that constraint permits.

The consequence should be stated plainly rather than discovered by the reader: **the precision to expect is fair, not high.** Several of the errors reported below are mesh-limited rather than model-limited, and we say which where we can tell them apart—helium's energy most clearly, where the softening moves the answer by nine percent and the quantities of physical interest are smaller than that. Where an h-ladder exists we give it instead of a single figure, and where the ladder has not flattened we say so. A finer mesh is the obvious next step and is not available under the constraint we have adopted.

Three properties of the determination of σ matter. It is performed on a *one-electron* system, where the model has no content of its own and the target is the exact model's own value. It is a single number at a given mesh, not a function of the system. And it is fixed before any two-electron system is computed. Everything below follows with no further calibration.

What this is not. It is not an optimisation of ε against a residual. That operation is available and must be refused: the total force residual is monotone in ε and minimised by unphysical settings—hydrogen ceases to bind at $\varepsilon = 0.75h$—so a scheme that tunes ε to improve an error measure drifts away from the physics it is meant to describe. Fixing it on a known energy is a different operation and has a unique answer.

3 The interface, and the two constructions

For a fixed partition, free variation of each density in (2) leaves a boundary term $\oint u_m \, \partial u_m/\partial n$, so the natural condition is homogeneous Neumann, $\partial u/\partial n = 0$, on each domain's own boundary—nothing is imposed. Solved that way the amplitudes at the interface generically *disagree*, and the interface is then placed where they agree. Continuity is therefore not a second assumption about Ψ; it is the equation that determines Γ. Three conditions, one extra unknown.

Two implementations each supply one half of that scheme and take the other half from elsewhere.

(A) Sharp partition, strict Neumann. Labelled cells, with the kinetic flux through any face joining cells of different ownership *omitted at assembly*. Omitting a face *is* the homogeneous-Neumann condition—zero flux implemented by not adding a flux, with no extra equation. This imposes the condition exactly but requires Γ to be known, which for H_2 and He is supplied by symmetry. One trap is worth recording: taking a *submatrix* of the full stiffness instead retains the diagonal contributions of faces leading into the excluded region and so imposes Dirichlet, pushing density off the boundary and under-binding. Measured on a one-electron problem outside a core, 3.977 against 6.177 eV; the latter matches a one-dimensional radial solve to 2%.

(B) Diffuse partition, continuity enforced weakly. An occupancy w_m evolves on the local amplitude mismatch $c = u_m - \max_{n\neq m} u_n$ and converges where that mismatch vanishes, so this construction genuinely *finds* Γ—the only one of the two that does. Its Neumann condition is only natural, not imposed: varying $\frac{1}{2}\int w|\nabla u|^2$ makes $w\,\partial u/\partial n = 0$ stationary, but on a finite grid a face straddling the interface carries half flux rather than none. That half-flux is

an extra kinetic cost sitting exactly at the interface, and it is a first-order discretisation error, not a structural one: H_2's bond error falls $34.9 \to 28.3 \to 21.4\%$ at $h = 0.1395, 0.10, 0.08$ and is still falling.

One qualification on that ladder, since it is the paper's only direct evidence of convergence. It was taken at *fixed* $\sigma = 1$, which predates the calibration (4). At fixed σ the softening $\varepsilon = \sigma h$ shrinks with the mesh on its own, so the ladder measures resolution *together with* a changing effective softening, and not resolution at fixed calibration. Under the declared scheme σ is re-determined at each h so that $E_h(\mathrm{H}) = -\frac{1}{2}$ throughout, which is a different trajectory. The ladder therefore shows that the error is mesh-sensitive and falling, which is the claim made of it here, but it is not the ladder the declaration implies and redoing it under $\sigma(h)$ is outstanding.

The two are not a choice between theories. They are two incomplete implementations of one: (A) imposes the condition and is told where Γ is; (B) finds Γ and imposes the condition weakly. Combining them—binary faces on a *diffuse* occupancy—does not work and should not be attempted: it gives H_2 a binding energy of $+110\%$ and a dipole of 0.108–0.187 D where reflection symmetry requires identically zero, because closed faces make an amplitude jump at the interface free. The diagnostic is direct: the kinetic energy of the assembled field $\Psi = \sum_m w_m u_m$, which charges that jump, exceeds the per-domain sum by a factor of 8.07, so the discrete solution is not in H^1.

What separates the two versions, measured. At a symmetry plane the two coincide, so distinguishing them needs an asymmetry—and the cleanest way to get one is to make it a property of the *interface* rather than of the nuclei: H_2 at fixed D with the dividing plane *displaced* off-centre, each domain still carrying unit charge. Both kernels are then $Z = 1$, one σ serves both, and the only asymmetry is the one under test. Writing $\delta E = \frac{1}{2} k z_i^2$ for a displacement z_i,

h	k, continuity enforced	k, strict per-domain	ratio
0.250	0.7359	0.1296	5.68
0.125	0.7394	0.1472	5.02

The continuity stiffness is converged—0.5% across a halving—while the strict one is not, climbing 13.6% and still moving. That asymmetry is the sharper statement: the strict construction charges for a *discontinuity*, a mesh-scale object, so it has no well-defined interface stiffness in the continuum limit, whereas the continuity construction does. For an interface that must *move*—which is every molecule without a symmetry plane—that is a difference in kind rather than in accuracy.

4 H_2: two kernels, interface between them

Hydrogen fixed one constant and settled nothing else. H_2 is the same model carried to two electrons *without any further calibration*: the same σ, the same functional, the same $i \neq j$ Coulomb rule, and the measured separation as the only other input. Every number in this section is therefore a prediction in the strict sense, and that is the only reason the agreement below is worth reporting—a two-electron constant fitted here would have made it worthless.

It is also the mildest two-electron system available. Each kernel is $Z = 1$, so the softening calibrated on hydrogen applies unchanged to both and nothing is transferred from a different kernel charge; and the interface sits at the midplane by reflection symmetry, in the *tail* of both densities rather than near either nucleus. The extension admits exactly two forms, differing only in how the interface condition is imposed—in a hard version and a weak version one—and we report both.

Each version is reported against *its own* one-electron reference, computed in the same code and at the same softening—a requirement rather than a nicety, for the reason given below. So the weak version uses $E(\mathrm{H}) = -0.499854$ and the hard version -0.500227, each exact to within a few parts in 10^4 of the $-\frac{1}{2}$ that (4) calibrates to.

The weak version gives the binding energy. With σ fixed by (4), it comes out

$$E_{\mathrm{bind}} = 2E(\mathrm{H}) - E(\mathrm{H}_2) = 2(-0.499854) - (-1.160855) = 0.161147,$$

at $R = 1.5\,a_0$, the commensurate separation nearest the observed 1.40028 (1.401 itself is not commensurate at $h = 0.1$). The other commensurate neighbour is worse: at $R = 1.3\,a_0$, $E = -1.133798$ gives $E_{\mathrm{bind}} = 0.134090$, i.e. -23.1%. Binding therefore *improves* from 1.3 to 1.5, so the two separations bracket the observed bond length but *not* the model's own minimum, which lies beyond 1.5—consistent with the uncalibrated scan, which placed it at 1.8 ± 0.1.

Against D_0 and D_e both, because neither comparison is clean. Two dissociation energies have to be distinguished. D_e is the *depth of the potential well*: the energy to separate the molecule into its atoms starting from the bottom of the well, with the nuclei at rest at the equilibrium separation. D_0 is the energy to separate it starting from the lowest vibrational level, which lies above the well bottom by the zero-point energy E_{ZPE}, so

$$D_0 \;=\; D_e \;-\; E_{\mathrm{ZPE}}, \qquad \text{for } \mathrm{H}_2: \quad \underbrace{0.164564}_{D_0} \;=\; \underbrace{0.174470}_{D_e} \;-\; \underbrace{0.009906}_{E_{\mathrm{ZPE}}} \qquad (E_h). \tag{5}$$

A dissociation experiment starts from the lowest vibrational level and therefore measures D_0; D_e is obtained from it by adding back a zero-point energy that must come from elsewhere. The two differ by 5.6% of the quantity being tested, which is not a small fraction of it:

$$\frac{0.161147 - 0.164564}{0.164564} = \mathbf{-2.1}\% \;\; (\text{vs } D_0), \qquad \frac{0.161147 - 0.174470}{0.174470} = \mathbf{-7.6}\% \;\; (\text{vs } D_e).$$

Each has one thing right and one wrong. $D_0 = 0.164564$ is the *measured* dissociation energy, so comparing against it compares the model with an observation rather than with another calculation—but D_0 includes nuclear zero-point motion, and the model has none, so the two are not the same quantity. $D_e = 0.174470$ *is* the same quantity—a well depth, which is what E_{bind} computes at fixed nuclei—but it is not measured: it is D_0 plus a zero-point energy supplied from outside.

We therefore give both and let the 5.6-point spread stand as what it is: a physical effect the construction omits, quantified. It is worth adding that the model cannot close the gap from within. Turning a well depth into a dissociation energy requires a vibrational zero-point energy, hence a nuclear mass and a constant of action—and neither appears anywhere in (2). D_0 is not reachable from inside this construction, which is a sharper statement of its scope than a choice between the two references would have been.

The hard version gives the bond. At its own minimum it returns E_{bind} too large by $+\mathbf{59.6}\%$—far worse than the weak version—while placing the bond at $1.349\,a_0$, within $-\mathbf{3.7}\%$ of the measured 1.40028. So the two versions fall on opposite sides of the observed binding energy, and each is the accurate one for a different observable:

	E_{bind} vs D_e	bond vs R_e
weak version	$-\mathbf{7.7}\%$	$+28\%$
hard version	$+59.6\%$	$-\mathbf{3.7}\%$

Neither is better on both, and §4.1 sets out what that does and does not license.

The bond, which is the better-behaved observable. The separation at which each construction's energy is lowest is a prediction carrying no reference at all beyond the measured $R_e = 1.40028$. The hard version puts it at $1.349\,a_0$, -3.7%, from a parabola through three commensurate points about the minimum. The weak construction at $\sigma = 1$ gives $1.8 \pm 0.1\,a_0$ ($+28\%$), the uncertainty being the width of the band within which the energy differences fall below the $\pm 0.0018\,E_h$ initialisation scatter. At the calibrated $\sigma = 0.627$ its minimum has not yet been located: a sharper kernel deepens the well and should shorten the bond, so we expect it between those two figures, but we have one point and not a scan, and report the gap rather than interpolate it.

The ordering is worth noting because it is the opposite of the energy's. On the bond the sharp construction is the accurate one, within four percent; on the binding energy it is the inaccurate one, at $+59.6\%$ against the weak split's -7.7%. No single construction here is better on both.

Two implementations, complementary errors, and what that does and does not license. The two differ in *exactly one* respect—how the interface condition is imposed—and everything else about them is the same model, the same calibration and the same geometry. So the complementarity is not a coincidence of two unrelated methods: the weak construction's error is a half-flux localised *at the interface*, which for a diatomic is precisely where the bond length is determined, and it is therefore large in the bond and much smaller in the binding energy. That the pair gets one observable right each is a statement about where the deficiency lies, and it localises it to the interface treatment alone.

We are careful about what this licenses. Quoting the hard version's bond together with the weak version's binding as "the model's answer" would require a rule, settled in advance, for which construction to trust for which observable. For the bond there is such a rule: H_2's interface position is fixed by reflection symmetry, so the hard version's *only* handicap—that Γ must be supplied to it from outside—is inactive here, and its geometry is accordingly the one to quote. For the binding energy no corresponding argument is available: the weak construction's handicap is never inactive, so its -7.7% is not privileged over the hard $+59.6\%$ by anything beyond being nearer the measurement, and selecting it on that ground would be fitting. The pair therefore *diagnoses*; it does not predict both observables at once, and we do not present it as doing so.

A warning earned the hard way. The reference must be computed at the same σ as the molecule. Paired instead with an atom measured at $\sigma = 1$, the same H_2 run reports $E_{\mathrm{bind}} = 0.188011$, i.e. $+7.8\%$ and apparently *over*-binding—the sign inverted by a mismatched reference. Since σ sets the cusp resolution and hence every energy, a reference at a different σ is not a reference at all.

5 He: hard split and weak split

Helium is the same two-electron question asked under far worse conditions. The kernel is $Z = 2$, so the softening calibrated on hydrogen is being extrapolated—though not unreasonably: the one-electron ion He^+ at the same σ gives -2.053249 against the exact -2.0, only 2.66% deep, so the cusp itself is well handled. And the interface passes *through* the kernel rather than lying between two of them, so it cuts the density maximum: $\int_\Gamma u^2 dS = 1.681$ against H_2's 0.496, a factor of 3.4.

Helium is special: both electrons touch the kernel. In every molecule below, each electron has its own kernel and the interface lies *between* them, in the tail of both densities.

Helium has one kernel and two electrons, so a dividing surface necessarily passes *through* it and both domains touch it. The interface therefore cuts the density maximum rather than the tail: $\int_\Gamma u^2\, dS = 1.681$ against H_2's 0.496, a factor of 3.4. Helium is not a smaller molecule; it is a different configuration, and its two admissible constructions are not two approximations to one answer.

Two constructions, describing different things. The *homogeneous* treatment puts both electrons in one domain sharing one shape. For a two-electron closed shell that is exactly the restricted Hartree–Fock ansatz—one spatial shape, one Coulomb pair, which the $i \neq j$ rule supplies directly—so it should return the Hartree–Fock energy, and the shortfall against the exact value is the correlation energy that neither carries. The *split* treatment gives each electron its own territory, with Γ the plane through the kernel and the condition imposed exactly:

$$E(\mathrm{He}) = -3.026749 \quad \text{against} \quad -2.903724, \qquad \textbf{over-binding by } 4.2\%.$$

What the homogeneous treatment measures. At the softening calibrated on hydrogen it returns $E = -2.771193$, with the state clean on every available check—$\mu = 0.000\,\mathrm{D}$ as sphericity requires, the norm exact at 2.000, and the interface measure 3×10^{-4} since there is no interface. The term breakdown is the informative part. Against the Hartree–Fock values $T = 2.8617$, $V_{eK} = -6.7492$, $J = 1.0258$:

σ	T	V_{eK}	V_{ee}
0.627 (calibrated on H)	2.5318 (−11.5%)	−6.3228 (−6.3%)	**1.0197** (**−0.6**%)
0.2846	3.1780 (+11.0%)	−7.3139 (+8.4%)	1.1135 (+8.6%)

At the calibrated σ the *pair* term is right to 0.6% and the whole error lies in the one-electron terms—which is where He^+ already placed it, with a 5.7% kinetic deficit at the same σ. So the two-electron content is sound and the unresolved $Z = 2$ cusp at $h = 0.1$ is what the energy is reporting.

And at this mesh the softening dominates helium. Moving σ from 0.627 to 0.2846 moves E by $0.251\,E_h$, about 9%, from 3.2% shallow of Hartree–Fock to 5.6% deep of it. Against that, the correlation energy is 1.45% of the total and the difference between the two constructions is smaller still. Three consequences follow and we state them rather than work around them. Helium's energy at $h = 0.1$ is a statement about resolution, not about the model. The gap to the exact value cannot be read as correlation, because the resolution error exceeds the quantity it would explain. And we decline to choose the σ that lands on Hartree–Fock: σ is fixed by hydrogen, and re-fixing it on helium would be the one thing this paper is organised to avoid.

A cross-check worth recording. At $\sigma = 0.2846$ the homogeneous treatment gives -3.022386 and the split treatment -3.026749—0.15% apart, from two different codes and two different constructions. That is a useful validation of both implementations, and it also shows that the distinction between them does *not* appear in helium's energy. Which is the point of the next paragraph: the difference is structural.

The split is worth keeping despite over-binding, and not as a variant to be averaged with the other. A spherical shared shape has no internal structure to orient, so it can generate no attraction between two such atoms at all; two territories can. Whether that mechanism accounts for the observed weak binding of He_2 we do not compute here, and we are careful not to claim it—dispersion-bound systems have been treated as outside this construction's range—but the *qualitative* difference is structural rather than numerical: one construction admits the

mechanism and the other forbids it. That is a reason to retain both for helium, and a reason not to read the two energies as a bracket around a single answer.

What helium establishes about the softening. The one-electron ion on the same kernel at the same σ gives -2.053249 against the exact -2.0, 2.66% deep, with $T = 1.8855$ against 2.0. So the $Z = 2$ cusp and the transferred softening are *both* sound, and whatever limits helium is a two-electron matter. That is worth having: it rules out the calibration and the kernel as explanations before any two-electron number is in hand.

And the geometry is the reason to expect helium to be the harder case either way. Its interface passes through the kernel rather than between two of them, so it cuts the density maximum: $\int_\Gamma u^2 dS = 1.681$ against H_2's 0.496. Whatever error an interface treatment carries is weighted there by the largest density in the system.

6 What this establishes, and what it does not

The formulation question is open, and the obvious test does not work. Which of the two versions is right—whether Γ should be placed by amplitude matching or the Neumann condition imposed on a Γ obtained otherwise—cannot be settled by either system here, since at a mirror plane the two coincide. The smallest asymmetric two-electron system is HeH^+, and we attempted it. It does not serve, for two independent reasons. Its rows do not converge: at the calibrated softening and separations 1.3 and $1.5\,a_0$ the total force residual is 121% and 163%, and since HeH^+ is heteronuclear that residual is not symmetry-cancelled but a genuine measurement, which fails by the same standard applied in §4. And the hard version cannot be run on it at all: the sharp implementation obtains the partner's potential by reflection, which requires a mirror plane, so there is no second version to compare against. A sharp solver that works without symmetry is the missing piece, and until it exists the question stands open.

Not that the model is accurate at two electrons. It is not. On H_2 the two forms disagree by sixty-odd points of D_e, and helium's two admissible treatments differ by more than the correlation energy they are being judged against. That width is the expected consequence of a declared property of the construction—the partition makes overlap impossible while the Coulomb term alone resists compression, and between equivalent domains across a mirror plane the partition contributes nothing—rather than a numerical failing to be driven down. The testable claim is that the error is *systematic*: one sign, one scale, and cancelling in differences taken at fixed composition.

And not that either observable has been predicted twice. One form places H_2's bond within 3.7% and the other its binding energy within 7.7%, from a single calibration performed on one electron. That is the paper's positive result, and its limit is stated in §4: the pair localises the deficiency to the interface treatment, and only the bond comes with a reason—H_2's Γ being fixed by symmetry—for preferring the form that gets it right.

A caveat on the references. The figure -1.174470 for H_2 is itself a *computed* quantity rather than a measured one, and the total electronic energy is in any case not an observable. The observables here are the dissociation energy, the bond length, and a dipole that must vanish by symmetry. We report total energies because they are what the two constructions differ in, not because they are measurable.

What the dipole buys. For a symmetric system the computed dipole must be identically zero, so any nonzero value is a pure error measurement needing no reference of any kind. It

is the one check in this paper that cannot be argued with, and it is what disqualified the combined construction of §3 and would disqualify any future variant that improved the energy while losing the symmetry.

References

[1] The simulator runs in any WebGPU browser at
https://claes542.github.io/RealMolecule/molecule_simulator.html
with source at https://github.com/Claes542/RealMolecule. Every figure in this paper is reproduced by a query string; for instance the H_2 row of §4 is
`?mol=h2&sep=1.5&box=12&h=0.10&soft=0.627&steps=233000&rc=0`
and the hydrogen calibration (4) is `?mol=h&box=20&h=0.10&soft=0.627`.

[2] C. Johnson, *RealQM Molecule Simulator: Validation on Small Molecules from Hydrogen and the Second Row*, companion article.